

Störmer's Rule

Mathematical Derivation from First Principles

Cosmos GNC Library — Propulsion Repository

2026

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1 Problem Statement

We consider a second-order ordinary differential equation of the form

$$y'' = f(x, y), \quad y(x_0) = y_0, \quad y'(x_0) = v_0, \quad (1)$$

where $y : \mathbb{R} \rightarrow \mathbb{R}^n$, $f : \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$, and crucially f *does not depend on* y' .

Remark 1 (Physical motivation). *Equation (1) describes any purely position-dependent force: gravitational n -body, Keplerian orbits, conservative mechanical systems, and the Störmer problem (charged particle in a magnetic field) that originally motivated the method. For the two-body orbital problem: $f(\mathbf{r}) = -\mu \mathbf{r}/|\mathbf{r}|^3$.*

Goal. Advance the solution from x_0 to $x_0 + H$ by subdividing H into M equal sub-steps of size $h = H/M$, in a way that:

1. exploits the absence of y' in f ,
2. produces an *even* error series in h (enabling Bulirsch-Stoer Richardson extrapolation), and
3. requires only $M/2$ new evaluations of f per column (half the cost of applying the standard modified midpoint rule to the equivalent first-order system).

2 The Störmer Three-Point Recurrence

2.1 Derivation from Taylor's Theorem

Write the Taylor expansions of y centered at $x_m = x_0 + mh$:

$$y(x_m + h) = y_m + hy'_m + \frac{h^2}{2}y''_m + \frac{h^3}{6}y'''_m + \frac{h^4}{24}y^{(4)}_m + \mathcal{O}(h^5), \quad (2)$$

$$y(x_m - h) = y_m - hy'_m + \frac{h^2}{2}y''_m - \frac{h^3}{6}y'''_m + \frac{h^4}{24}y^{(4)}_m + \mathcal{O}(h^5). \quad (3)$$

Adding (2) and (3) cancels all odd powers:

$$y(x_m + h) + y(x_m - h) = 2y_m + h^2y''_m + \frac{h^4}{12}y^{(4)}_m + \mathcal{O}(h^6). \quad (4)$$

Substituting $y''_m = f(x_m, y_m)$ and rearranging:

$$\boxed{y_{m+1} = 2y_m - y_{m-1} + h^2f(x_m, y_m) + \mathcal{O}(h^4).} \quad (5)$$

This is the **Störmer three-point formula**: given two successive solution values y_{m-1} and y_m , the next value y_{m+1} requires a single evaluation of f and is accurate to $\mathcal{O}(h^4)$ locally.

Remark 2 (No velocity in the recurrence). *Equation (5) uses y_{m-1} , y_m , and $f(x_m, y_m)$ only. The velocity y' has been eliminated algebraically. The price is the need for two starting values.*

2.2 The δ -Increment Formulation

Define the position increment $\delta_m := \theta_{m+1} - \theta_m$, the change in the numerical solution over one sub-step. From (5):

$$\begin{aligned} \delta_m &= y_{m+1} - y_m \\ &= (2y_m - y_{m-1} + h^2f_m) - y_m \\ &= (y_m - y_{m-1}) + h^2f_m \\ &= \delta_{m-1} + h^2f_m. \end{aligned} \quad (6)$$

This telescoping relation propagates δ forward with only h^2f_m added at each step. The velocity information is encoded in the initial δ_0 .

Starting value from Taylor: The value $y_1 = y(x_0 + h)$ from (2) with $m = 0$:

$$y_1 = y_0 + h v_0 + \frac{h^2}{2} f_0 + \mathcal{O}(h^3), \quad (7)$$

so

$$\boxed{\delta_0 = h v_0 + \frac{h^2}{2} f_0.} \quad (8)$$

3 Störmer's Modified Midpoint Algorithm

Notation. We introduce θ_m to denote the **numerical approximation** to the true solution $y(x_m)$ at sub-step m , where $x_m = x_0 + mh$. The exact solution is $y(x_m)$; its approximation is θ_m . The algorithm computes a sequence of approximations:

$$\theta_0 \approx y(x_0), \quad \theta_1 \approx y(x_1), \quad \dots, \quad \theta_M \approx y(x_0 + H).$$

The velocity approximation at the final point is denoted $z_M \approx y'(x_0 + H)$.

Why θ instead of y ? In numerical analysis, it is standard to distinguish the *exact* solution $y(x)$ (unknown, continuous) from the *computed* approximation θ_m (known, discrete). The error at step m is $y(x_m) - \theta_m$. This distinction becomes crucial when analyzing convergence and error propagation.

The complete algorithm for M sub-steps of size $h = H/M$ is:

1. **Initialize.**

$$\theta_0 = y_0, \quad (9)$$

$$\delta_0 = h v_0 + \frac{h^2}{2} f(x_0, \theta_0). \quad (10)$$

Set $\theta_1 = \theta_0 + \delta_0$.

2. **Sub-step loop** for $m = 1, \dots, M - 1$:

$$f_m = f(x_0 + mh, \theta_m), \quad (11)$$

$$\delta_m = \delta_{m-1} + h^2 f_m, \quad (12)$$

$$\theta_{m+1} = \theta_m + \delta_m. \quad (13)$$

3. **Final derivative** (velocity estimate at $x_0 + H$):

$$z_M = \frac{\delta_{M-1}}{h} + \frac{h}{2} f(x_0 + H, \theta_M). \quad (14)$$

The output is $(\theta_M, z_M) \approx (y(x_0 + H), y'(x_0 + H))$, the numerical approximations to the true position and velocity at $x_0 + H$, with local truncation error $\mathcal{O}(h^2)$.

Total function evaluations of f : one per sub-step $m = 0, \dots, M$, i.e. $M + 1$ total, but f_0 is supplied from the previous step, so M *new evaluations*.

4 Equivalence to Störmer-Verlet / Leapfrog

Define the half-integer velocity $v_{m+1/2} := \delta_m/h$. Rewriting the algorithm in terms of v :

$$v_{1/2} = \frac{\delta_0}{h} = v_0 + \frac{h}{2} f_0 \quad (\text{half-kick}), \quad (15)$$

$$\theta_{m+1} = \theta_m + h v_{m+1/2} \quad (\text{drift}), \quad (16)$$

$$v_{m+3/2} = v_{m+1/2} + h f_{m+1} \quad (\text{full kick, } m = 0, \dots, M-2), \quad (17)$$

and the final velocity:

$$z_M = v_{M-1/2} + \frac{h}{2} f_M. \quad (18)$$

Equation (18) can be read as: take the half-integer velocity $v_{M-1/2}$, and kick it forward by a half step to return to integer time M . Equivalently:

$$z_M = \frac{v_{M-1/2} + v_{M+1/2}}{2}, \quad \text{where } v_{M+1/2} = v_{M-1/2} + h f_M. \quad (19)$$

So z_M is the arithmetic mean of the bracketing half-integer velocities, a second-order approximation to $y'(x_0 + H)$.

Remark 3 (Symplecticity). *The drift-kick-drift structure (15)–(17) is the Störmer-Verlet integrator, which is symplectic for Hamiltonian systems $H = \frac{1}{2}|v|^2 + V(y)$ with $f = -\nabla V$. It preserves a modified Hamiltonian exactly to all orders. Importantly, it does NOT suffer from secular energy drift.*

5 Even Error Series: Gragg's Theorem

Theorem 1 (Gragg 1965). *Let $y(x)$ be sufficiently smooth. For Störmer's modified mid-point rule applied to $y'' = f(x, y)$ with M sub-steps of $h = H/M$ and the starting/ending conventions (8)–(14), the global errors admit expansions in even powers of h only:*

$$y(x_0 + H) - \theta_M = a_2(H) h^2 + a_4(H) h^4 + a_6(H) h^6 + \dots, \quad (20)$$

$$y'(x_0 + H) - z_M = b_2(H) h^2 + b_4(H) h^4 + b_6(H) h^6 + \dots, \quad (21)$$

where the coefficients $a_{2k}(H)$, $b_{2k}(H)$ are independent of M .

Sketch of proof (time-reversibility argument). The Störmer recurrence (9)–(14) is *time-reversible*: reversing $h \rightarrow -h$ and the sequence of θ_m recovers the original sequence. A time-reversible method cannot accumulate odd powers of h in its global error, because odd-power terms change sign under $h \rightarrow -h$ while the exact solution does not. More formally, one expands $\theta_m(h)$ as a power series in h and shows that all odd-power terms cancel identically (see Hairer, Nørsett & Wanner, Vol. I, §II.8 for the complete argument).

Corollary 1 (Richardson extrapolation efficiency). *Because the error series (20) contains only $h^2, h^4, \dots$, a single Richardson extrapolation step using results at h and $h/2$ eliminates the h^2 term and yields $\mathcal{O}(h^4)$ accuracy. In general, k extrapolation steps yield $\mathcal{O}(h^{2k+2})$ accuracy from $\mathcal{O}(h^2)$ base integrations. This is the foundation of the Bulirsch-Stoer method.*

6 Richardson Extrapolation Setup (Bulirsch-Stoer)

Evaluate the modified midpoint rule with sub-step counts $M_1 < M_2 < M_3 < \dots$ (the Bulirsch-Stoer sequence: $\{2, 4, 6, 8, 12, 16, \dots\}$), giving approximations $T_{j,0} := \theta_{M_j}(H)$.

Define the extrapolation table recursively:

$$T_{j,k} = T_{j,k-1} + \frac{T_{j,k-1} - T_{j-1,k-1}}{(M_j/M_{j-k})^2 - 1}, \quad k = 1, 2, \dots, j. \quad (22)$$

Each extrapolation step eliminates the next order in h^2 . The diagonal $T_{k,k}$ converges to $y(x_0 + H)$ as k increases.

Remark 4 (Computational cost vs. accuracy). *For the first-order system $[y', f(x, y)]^T$ of dimension $2n$, the modified midpoint rule needs M evaluations of f per column. Störmer's rule needs only $M/2$ evaluations per column (the NR3 code uses $n_{\text{eff}} = M/2$ leapfrog steps of $h_{\text{eff}} = 2h = 2H/M$, as shown below) for the same total interval H . Since f can be expensive (e.g., gravitational force with many bodies), this factor of two is significant.*

7 Analysis of the NR3 Implementation

7.1 Parameterization

NR3's `StepperStoerm::dy()` uses:

- `nstep` = M (from Bulirsch-Stoer sequence $\{2, 4, 6, 8, \dots\}$).
- `h` = H/M (unit sub-step).
- `h2` = $2h = 2H/M$ (actual loop step, double the unit sub-step).
- `nstp2` = $M/2$ (number of loop iterations).

7.2 Algorithm in NR3 variables

Initialization:

$$\begin{aligned} \text{ytemp}[i] &\leftarrow y[i] & \theta_0 &= y_0, \\ \text{ytemp}[n+i] &\leftarrow y'[i] + h \cdot f_0 & v^* &= v_0 + h f_0 = v_{1/2} \cdot \underbrace{2}_{h_{\text{eff}}/h}. \end{aligned}$$

In other words, the stored “velocity” `ytemp` $[n+i]$ is the half-integer velocity $v_{1/2} = v_0 + (h/2) f_0$ scaled by 2: `ytemp` $[n+i] = 2 v_{1/2}$. Then the drift step below with factor $h_{\text{eff}} = 2h$ is:

$$\theta_1 = \theta_0 + h_{\text{eff}} \cdot \text{ytemp}[n+i]/2 = \theta_0 + h_{\text{eff}} v_{1/2} = \theta_0 + 2h (v_0 + \frac{h}{2} f_0) = \theta_0 + 2h v_0 + h^2 f_0.$$

Alternatively, viewing the NR3 scheme directly as Störmer-Verlet with step $h_{\text{eff}} = 2h$:

$$\text{Loop } m = 1, \dots, n_{\text{eff}} : \quad \begin{cases} \theta_m = \theta_{m-1} + h_{\text{eff}} v_{m-1/2}, \\ v_{m+1/2} = v_{m-1/2} + h_{\text{eff}} f(x_0 + m h_{\text{eff}}, \theta_m), \end{cases} \quad m < n_{\text{eff}}. \quad (23)$$

Final velocity: $z = v_{n_{\text{eff}}-1/2} + (h_{\text{eff}}/2) f(x_0 + H, \theta_{n_{\text{eff}}})$.

7.3 Cost comparison

Method	State dimension	Evals of f per column	Error order
Midpoint (first-order 2n-system)	$2n$	M	$\mathcal{O}(h^2)$
Störmer (NR3)	n	$M/2$	$\mathcal{O}(h_{\text{eff}}^2)$

Both methods embed in a Bulirsch-Stoer extrapolation with an even-power error series and converge to the same accuracy. Störmer uses half the evaluations of f at the cost of doubling the effective sub-step size, which the extrapolation corrects.

8 Is Inheritance Necessary?

NR3's design uses `struct StepperStoerm : StepperBS<D> solely` for code reuse of the Bulirsch-Stoer extrapolation machinery (odeint driver, column sequence, Richardson table, stepsize control). The method `dy()` is entirely independent of the base class algorithms; it overrides only the *sub-step integration* while reusing everything else.

From first principles, the minimal reusable components are:

1. **StormerStep**: executes n_{eff} leapfrog sub-steps over a given interval H . No base class required. Template on `DerivativeType` and `ContainerT`.
2. **Error estimation**: step-doubling (compare n_{eff} steps vs. $2n_{\text{eff}}$ steps) or Richardson extrapolation as a separate, composable component.
3. **Step-size control**: compose with the existing `ComputePISetpSize / PISetpSizeControl` classes already present in the library.

Composition over inheritance, exactly consistent with how `CalculateNewYAndError` and `IntegrateWithPIControl` were designed.

9 Harmonic Oscillator Test Case

The standard test for Störmer's rule is the harmonic oscillator:

$$y'' = -\omega^2 y, \quad y(0) = 1, \quad y'(0) = 0, \quad \omega = 1. \quad (24)$$

Exact solution: $y(t) = \cos t$, $y'(t) = -\sin t$, with conserved energy:

$$E(t) = \frac{1}{2} (y'(t))^2 + \frac{1}{2} (y(t))^2 = \frac{1}{2}. \quad (25)$$

Störmer-Verlet (symplectic) conserves a modified energy exactly, so the energy error remains bounded for all time (no secular drift), unlike non-symplectic methods.

Expected convergence. For a single step of size $H = 0.5$ with n_{eff} leapfrog sub-steps of $h_{\text{eff}} = H/n_{\text{eff}}$:

$$|y(H) - \theta_M| \approx \frac{H^3}{24} |y'''(0)| \cdot h_{\text{eff}}^2 = \frac{H^3 \omega^3}{24} \cdot h_{\text{eff}}^2 = \mathcal{O}(h_{\text{eff}}^2).$$

Doubling n_{eff} (halving h_{eff}) should reduce the error by a factor of 4. This is the convergence test in the unit tests.

10 Summary

1. Störmer’s rule exploits $y'' = f(x, y)$ (no y' in f) via the three-point recurrence $y_{m+1} = 2y_m - y_{m-1} + h^2 f_m$.
2. In δ -form: $\delta_m = \delta_{m-1} + h^2 f_m$, with starting value $\delta_0 = hv_0 + \frac{h^2}{2} f_0$.
3. The algorithm is equivalent to Störmer-Verlet (leapfrog), which is symplectic for Hamiltonian systems.
4. Gragg’s theorem guarantees an even-power error series in h , enabling efficient Bulirsch-Stoer Richardson extrapolation.
5. NR3’s `StepperStoerm` inherits from `StepperBS<D>` for code reuse only. The core algorithm (`dy()`) is independent and should be implemented as a standalone class.
6. For M Bulirsch-Stoer sub-steps, Störmer uses $M/2$ evaluations of f vs. M for the first-order midpoint rule applied to the $2n$ -dimensional system — a factor-of-two saving.

References.

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